

Name: Diksha Utekar
Devansh Wadhwani

Roll no.: 63/64

WT EXPERIMENT - 4

Aim: Component and Arduino Introduction

Introduction:

The Arduino Uno is a versatile and widely used microcontroller board, forming the backbone for countless electronics projects. In this experiment, we will use an Arduino Uno to build a water turbidity detector that can assess the clarity of water using a turbidity sensor. The system helps monitor water quality in real-time, with applications in environmental monitoring, water treatment plants, and aquaculture.

Arduino Uno Overview

The Arduino Uno is one of the most popular boards in the Arduino family due to its simplicity, low cost, and ease of use. It features:

- **Microcontroller:** Atmega328P
- **Operating Voltage:** 5V
- **Digital I/O Pins:** 14 (6 can be used as PWM outputs)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB
- **Clock Speed:** 16 MHz
- **USB Connection:** For programming and communication
- **Power Supply:** Can be powered via USB or an external power supply

The Arduino Uno can be programmed using the Arduino IDE, which allows users to write and upload code that controls various electronic components connected to the board.

Installation and Setup

Installing the Arduino IDE

1. Download and install the Arduino IDE from the official website.
2. Connect the Arduino Uno via USB.
3. In the IDE, go to Tools > Board > Arduino Uno.
4. Select the appropriate COM port under Tools > Port.
5. Upload a sample sketch to verify functionality.

Running a Test Blink Program

```
void setup() {  
  pinMode(LED_BUILTIN, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(LED_BUILTIN, HIGH);  
  delay(1000);  
  digitalWrite(LED_BUILTIN, LOW);  
  delay(1000);  
}
```

If the built-in LED blinks, your setup is working fine.

Components Used

1. **Arduino Uno** – Central microcontroller
2. **Turbidity Sensor (e.g., SEN0189)** – Measures water clarity
3. **LCD** – Alerts for high turbidity and or showing turbidity levels
4. **Resistors, Jumper Wires, Breadboard** – For connections

Code:

```
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 2, 16);

int sensorPin = A0;

void setup()
{
  Serial.begin(9600);
  // initialize LCD
  lcd.init();
  // turn on LCD backlight
  lcd.backlight();
}

void loop() {
  int sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  int turbidity = map(sensorValue, 0, 520, 100, 0);
  delay(100);
  lcd.setCursor(0, 0);
  lcd.print("Turbidity:");
  lcd.print(" ");
  lcd.setCursor(10, 0);
  lcd.print(turbidity);

  delay(100);
  if (turbidity < 20) {
    lcd.setCursor(0, 1);
    lcd.print(" its CLEAR ");
    Serial.print(" its CLEAR ");
  }
  if ((turbidity > 20) && (turbidity < 50)) {
    lcd.setCursor(0, 1);
    lcd.print(" its CLOUDY ");
    Serial.print(" its CLOUDY ");
  }
  if (turbidity > 50) {
```

```
    lcd.setCursor(0, 1);  
    lcd.print(" its DIRTY ");  
    Serial.print(" its DIRTY ");  
  }  
}
```

Code Breakdown

Libraries

- `#include <LiquidCrystal.h>`: Includes the LiquidCrystal library to control the LCD display (used for displaying turbidity levels and alerts).

Pin Setup

- **Turbidity Sensor**: turbidityPin (A0)
- **LCD Display**: LiquidCrystal(rs, en, d4, d5, d6, d7)
 - Example pin configuration: rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2

Setup Function

- Initializes the **serial monitor** for data logging.
- Configures the **analog input** pin to read turbidity sensor values.
- Initializes the **LCD** and displays a welcome or standby message.

Turbidity Reading Function:

- Reads the **analog value** from the turbidity sensor.
- **Maps or compares** it to a predefined threshold.
- Returns the raw or processed turbidity level for further action.

Methodology:

1. Turbidity Sensing:

The turbidity sensor continuously measures the water clarity. Higher values indicate murkier water (more suspended particles).

2. Arduino Processing:

The sensor value is read and processed by Arduino. If the turbidity crosses a set threshold, a warning is activated using an LED or buzzer.

3. Monitoring & Feedback:

Real-time values are printed on the Serial Monitor, helping in calibration and analysis.

Conclusion:

This experiment demonstrates how an Arduino Uno can be used to monitor water quality by detecting turbidity. The project has practical applications in water treatment systems, environmental studies, and smart agriculture. With further development, it could be integrated into IoT platforms for remote monitoring and alerts.